# Session 4 (Live demo): the 2×2 calculator

**Foundations of Clinical Research** · Week 4 · Saturday 19:30-22:30, online · Neudata · Instructor script

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|---|---|
| **Slots** | **Main demo (8 min, 20:37-20:45):** section 4.2, just before the break, slide 27 "Live demo: the 2x2 calculator" (sheet *Association*). **Checking the class calculations (15 min, 21:00-21:15):** slide 29, after the break. **Short add-on (1-2 min, optional):** section 4.3, slide 32 "Same test, different setting: PPV depends on prevalence" (sheet *Diagnostic*). |
| **File** | `2x2_calculator.xlsx` (regenerate with `python make_2x2_calculator.py`) |
| **Software** | Excel or LibreOffice Calc (any recent version), shown to the class by sharing your screen in Zoom or Teams. |
| **Participants** | Watch the shared screen; they answer the calculation polls after the break. The file is emailed to them after the session. |
| **Goal** | Everyone sees risk, RR, RD, NNT and OR computed from ONE table and hears each one in plain words. In 4.3 they see the same test give very different PPVs in two settings. |

## Before the session (5 minutes, the day before)

1. Open the file and check that it shows the **worked example**: Restrictive 20 / 80, Usual care 30 / 70 (sheet **Association**); 18 / 16 / 2 / 64 and a new-setting prevalence of **2%** (sheet **Diagnostic**). If someone has changed it, rerun `make_2x2_calculator.py`.
2. Set the zoom to about **150%** so it is readable on participants' laptops and phones. Close every other window and mute notifications before you share the screen.
3. Only the yellow cells can be edited. The sheets are protected without a password (Review, then Unprotect Sheet if you need to change a label).
4. Have a shared online document open as a fallback (see the end of this script).

## Part 1: Main demo in 4.2 (8 minutes, sheet *Association*)

### Step 1: Show the layout (1 minute)

**Do:** share your screen and open sheet **Association**. Point at the yellow cells with the cursor.

**Say:** *"Same layout as the slide: treatment or exposure in rows, with the treated group on top, and outcome in columns, with 'died' on the left. a, b, c and d are always in the same place."*

**Appears:** the 2×2 table with 20 | 80 and 30 | 70, totals 100 and 100.

### Step 2: Read the results in words (3 minutes)

**Do:** point with the cursor at each row of the results table, reading the right-hand "In plain words" column aloud.

| Row | Value shown | Plain words shown | Teaching point |
|---|---|---|---|
| Risk in exposed / treated | 20.0% | 20 of 100 patients in the Restrictive group had the outcome | Risk is out of everyone in the group. |
| Risk in unexposed / control | 30.0% | 30 of 100 ... Usual care ... | |
| Risk ratio (RR) | 0.67 | The risk is 33% LOWER in the exposed/treated group | Relative effect. |
| Risk difference (RD) | −10.0% | 10.0 fewer patients with the outcome per 100 | Absolute effect: the one patients need. |
| NNT / NNH | 10.0 | NNT: treat about 10 patients to PREVENT one outcome | |
| Odds (two rows) | 0.25 and 0.43 | Odds compare those WITH the outcome to those WITHOUT | Odds are not risks. |
| Odds ratio (OR) | 0.58 | outcome is common, so OR lies further from 1 than RR | Matches the OR-vs-RR chart. |

## Step 3: Make the outcome rare (2 minutes)

**Do:** type **2** in a (Restrictive, died), **98** in b, **3** in c (Usual care, died), **97** in d. Press Enter after each.

**Appears:**

| | Before | After |
|---|---|---|
| Risks | 20% vs 30% | 2% vs 3% |
| RR | 0.67 | **0.67 (unchanged)** |
| RD | −10% | **−1%** |
| NNT | 10 | **100** |
| OR | 0.58 | **0.66, now close to RR** |

**Say:** *"Same relative effect, but now we must treat 100 patients to prevent one death. And because death is now rare, the OR has almost caught up with the RR."*

**Do:** type the original numbers back (20, 80, 30, 70).

## Step 4: Case-control warning (1 minute)

**Do:** point with the cursor at the red line under the table.

**Say:** *"The spreadsheet will happily calculate a 'risk' from case-control data. Do not read it, because the researchers chose how many cases and controls to include. In the second calculation after the break, only the OR is valid. After the break we will do two calculations together: Part C of your Session 4 exercise sheet."*

## Part 2: Add-on in 4.3 (1-2 minutes, optional, sheet *Diagnostic*)

Use this only if the file is already open and you are on time. The slide makes the same point without it.

1. **Do:** share your screen and click the **Diagnostic** tab. **Appears:** sensitivity 90.0%, specificity 80.0%, PPV 52.9%, NPV 97.0%, prevalence 20.0% (the ICU picture).
2. **Do:** point with the cursor at the blue box *"Same test, another setting"*. The yellow prevalence cell shows **2%**. **Appears:** PPV in the new setting **8.4%**, NPV **99.7%**.
3. **Say:** *"Same sensitivity, same specificity. Screening a population where 2 in 100 have the disease, fewer than 1 in 10 positive results are real."*
4. **Do (optional):** point with the cursor at the small table underneath: prevalence 1% / 2% / 5% / 20% / 50% gives PPV 4.3% / 8.4% / 19.1% / 52.9% / 81.8%.

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## Teaching points to land

1. **One layout, every measure.** Most errors are layout errors, not arithmetic errors.
2. **Relative versus absolute.** An RR of 0.67 can mean an NNT of 10 or of 100, depending on the baseline risk.
3. **OR ≈ RR only when the outcome is rare.**
4. **Case-control data: OR only.**
5. **Sensitivity and specificity belong to the test. PPV and NPV depend on how common the disease is.**

## After the break: checking the class calculations (15 minutes, slide 29)

Share the screen again. For each calculation, run the polls first (see s4_solutions, Part C), then type the numbers into the Association sheet so everyone sees the answers:

| Exercise (exercise sheet Part C) | Type | Shows |
|---|---|---|
| Calculation 1 (fluid trial, 200 per group) | a 20, b 180, c 30, d 170 | RR 0.67, RD −5.0%, NNT 20, OR 0.63 |
| Calculation 2 (fluid case-control, new dialysis) | a 24, b 16, c 16, d 64 | OR **6.00** (ignore the "risks") |

Ask a volunteer to say each result in one plain sentence, including why Calculation 2 gives only an OR (and whether the arrow could run the other way). Then restore the worked example (20, 80, 30, 70) so the file is ready for the next course.

## Fallback: screen sharing or the spreadsheet fails

Type this table into the chat or a shared online document and work through it with the class. The numbers are chosen to be easy:

| | Died | Survived | Total | Risk |
|---|---|---|---|---|
| Restrictive | 20 | 80 | 100 | 20/100 = 0.20 |

| | Died | Survived | Total | Risk |
|---|---|---|---|---|
| Usual care | 30 | 70 | 100 | 30/100 = 0.30 |

- RR = 0.20 / 0.30 = 0.67
- RD = 0.20 − 0.30 = −0.10 (10 fewer deaths per 100)
- NNT = 1 / 0.10 = 10
- OR = (20 × 70) / (80 × 30) = 1400 / 2400 = 0.58

For the rare-outcome point, use 2/100 vs 3/100: RR = 0.67, NNT = 100, OR = (2 × 97) / (98 × 3) = 194 / 294 = 0.66.

For the PPV point, use the slide "Same test, different setting" (PPV 90/170 = 53% in the ICU; 9/107 = 8% in screening).

## Common problems

| Problem | Fix |
|---|---|
| Cells show − | One of the four counts is empty or zero in a denominator. Type a number in every yellow cell. |
| Cannot type in a cell | Only yellow cells are editable (protected sheet). Review, then Unprotect Sheet (no password). |
| Formulas show as text (LibreOffice) | Tools, Options, LibreOffice Calc, View: untick "Formulas"; or press Ctrl + `. |
| Prevalence typed as 2 instead of 2% | Type `2%` or `0.02`. The cell is formatted as a percentage. |
| Numbers look wrong | Check the layout: treated/exposed on top, "died" on the left. |